Hydraulic Bayesian mortality model manual

Version 2 theory and technical guide with xylem growth and optional repair

3 November 2026

This manual explains how the model predicts stem water potential, accumulates hydraulic stress exposure and estimates mortality probability. It covers the scientific equations, their provenance, all input parameters, the Linux workflow, Bayesian calibration and interpretation of the outputs.

Code baseline. This guide describes version 2.0 of hydraulic_bayesian_mortality_F95.zip, updated on 3 November 2026. Version 2 adds potential and functional xylem conductance, seasonal growth, optional gated repair, optional proportional turnover and a prescribed seasonal leaf-area factor. It retains the Bayesian survival formulation. All supplied environmental scenarios, prior values and synthetic death counts are illustrative.

Scientific status. The hydraulic equations use established transport and storage principles. The specific relationship between hydraulic exposure and mortality hazard is a proposed model to calibrate and test. Crossing P50 does not establish that a tree will die.

Verification status. Version 2 was compiled with GNU Fortran 13.3.0 using strict F95 and runtime checks. The executable passed analytic capacity-process tests, water-balance and P50-consistency checks, Bayesian likelihood checks, backward-compatibility and timestep checks. See Section 14 and the packaged verification record for details. These checks establish numerical behaviour, not empirical mortality validity.

Guide to the manual

Sections 1 to 7 explain scope, hydraulic theory, stomatal regulation, embolism, mortality, Bayesian inference and numerical integration. Sections 8 to 14 cover installation, drivers and cases, parameter files, calibration, outputs and verification. Section 15 contains the references and their relevance.

Version 2 keeps the existing five-column drivers usable. Seven-column drivers additionally prescribe seasonal growth activity and relative leaf area. Repair is disabled by default. All new rates and eligibility thresholds are hypotheses for calibration, not literature-derived species constants.

The Word version contains editable mathematical equations. The PDF provides a fixed-layout reading copy. Mathematical symbols are mapped to the exact Fortran parameter names in the technical tables.

1 Model scope and conventions

What the model represents

The tree is represented by a single water-storage compartment connected to a prescribed root-zone soil potential. Water enters through a series hydraulic resistance and leaves through transpiration. New functional xylem and optional conduit repair can restore capacity; proportional turnover can remove retained xylem. Stem potential controls stomatal closure and xylem damage; damage increases hydraulic resistance and therefore changes subsequent stem potential.

The mortality calculation follows this hypothetical living-tree trajectory. It integrates an instantaneous hazard and converts the accumulated hazard to survival probability. The program does not randomly kill trees during the hydraulic simulation, stop transpiration after a sampled death or simulate decomposition.

One run represents a shared species or parameter group. Cases can have different environmental time series and follow-up durations, but share base hydraulic parameters, initial states and mortality parameters. Growth activity and relative leaf area can differ through their forcing records. To compare species, use separate run directories with species-specific parameters. There is no hierarchical species model, block effect or individual frailty term.

Units and signs

Quantity	Symbol	Units and convention
Stem and soil potential	Ψ, Ψ_s	MPa; tension is negative
Hydraulic flow and transpiration	Q, E	L h^{-1}; uptake is signed
Hydraulic resistance	R	MPa h L^{-1}
Capacitance	C	L MPa^{-1}
Hydraulic time	t_h	Hours
Mortality elapsed time	t_d	Days since the first forcing row
Fractional conductance loss	L	0 to 1; PLC = 100 L
Mortality hazard coefficients	λ_0, k	Day^{-1}

For negative potentials, a stem potential of −4 MPa is more stressful than −2 MPa. If P50 is −3 MPa, −4 MPa is below P50 and gives positive potential-based exposure. Avoid the ambiguous phrase “exceeds P50” without defining the sign convention.

Required evidence

Environmental drivers are temperature, root-zone volumetric moisture, VPD and an externally prepared daylight or light-response factor. Independent measurements should constrain leaf area, soil retention, stomatal response, resistance, capacitance and vulnerability width. Absolute mortality calibration additionally requires observed survival/death outcomes. Hydraulic measurements alone do not identify an absolute mortality probability.

2 Hydraulic transport and storage

Electrical circuit analogy

Water potential corresponds to voltage, flow to electrical current, resistance to resistance and stored water to capacitance. The framework follows compartment water-balance models such as Ruffault et al. (2022) and the whole-plant RC approach discussed by Zhang et al. (2025). Our implementation reduces the plant to one effective stem compartment.

The soil-to-stem inflow is

$$Q = \frac{\Psi_s - \Psi - \Psi_g}{R}, \Psi_g = 0.00981\, h. \quad (1)$$

Here h is the effective height above the soil in metres. The gravity term has units MPa. It can be small for seedlings but is retained in the code. If the stem is sufficiently wetter than the effective soil source, Q can become negative; the model permits reverse exchange with its single soil compartment.

The storage balance gives

$$C \frac{d\Psi}{dt_h} = Q - E. \quad (2)$$

Capacitance is positive. Transpiration greater than uptake depletes water storage and makes stem potential more negative. Greater capacitance slows this response. For fixed resistance, the RC response time is $\tau = RC$, in hours. This is a response timescale, not a mortality threshold.

Converting soil moisture to soil potential

The inverse van Genuchten (1980) retention relation is used:

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, m = 1 - \frac{1}{n}. \quad (3)$$

$$\Psi_s = -\frac{0.00981}{\alpha}\left(S_e^{-1/m} - 1\right)^{1/n}. \quad (4)$$

The moisture input θ is volumetric water content in $m^3\ m^{-3}$. The parameter α is in m^{-1}, so the resulting pressure head is converted to MPa using 0.00981 MPa per metre of water head. The input must satisfy $\theta_r < \theta \leq \theta_s$; the code rejects values outside this interval.

Fit the retention parameters to the soil in the experiment. The code represents matric potential and omits osmotic potential. Because moisture is prescribed, rainfall, drainage and soil-water depletion are not simulated. A measured-curve interpolation could replace soil_potential, but this requires a code change.

3 Stomatal regulation and transpiration

Effective conductance

The code uses an empirical multiplicative response, following the general approach of Jarvis (1976). Coupling stomata to water potential is also motivated by Tuzet et al. (2003). The exact functions below are chosen parameterisations, not a reproduction of either full model.

$$g_w = g_{min} + (g_{max} - g_{min}) f_{day} f_T f_D f_\Psi . \quad (5)$$

$$f_T = \exp\left[-\left(\frac{T - T_{opt}}{w_T} \right)^2 \right], f_D = \frac{1}{1 + D / D_0} . \quad (6)$$

$$f_\Psi = \frac{1}{1 + \exp\left[(\Psi_{gs50} - \Psi) / s_g \right]} . \quad (7)$$

Temperature T is in degrees Celsius; VPD D and its response scale D_0 are in kPa. The water-potential width s_g must be positive. At $\Psi = \Psi_{gs50}$, the stress-sensitive response f_Ψ equals 0.5. This does not generally mean that total conductance equals $g_{max}/2$, because residual conductance and other environmental factors remain in Equation 5.

The externally supplied f_{day} lies between 0 and 1. At night, setting it to zero leaves $g_w = g_{min}$. There is no explicit stomatal time lag or independent stomatal damage memory. Temperature already contributes to measured VPD; f_T describes an additional empirical direct response, so it should not duplicate a saturation-vapour-pressure multiplier.

Whole-tree transpiration

The vapour-transfer approximation, consistent with the biophysical framework described by Campbell and Norman (1998), is

$$E = 64.8 \, A_{L,0} f_A g_w \frac{D}{P_{atm}} . \quad (8)$$

Base leaf area $A_{L,0}$ is in m² and f_A is the prescribed leaf_area_factor (1 for legacy five-column inputs). Conductance is in mol m^{-2} s^{-1} and both pressures in kPa. The factor 64.8 converts mol s^{-1} water to approximately L h^{-1}, using a water molar mass of about 0.018 kg mol^{-1} and density of about 1 kg L^{-1}.

Air temperature is assumed to equal leaf temperature, and effective vapour conductance is approximated by stomatal conductance. Boundary-layer resistance and leaf energy balance are omitted. Consequently, the output gsw_mol_m2_s should be interpreted within these assumptions when compared with leaf gas-exchange measurements.

4 Embolism growth and hydraulic recovery

Vulnerability remains linked to P50

The sigmoidal vulnerability curve is retained:

$$L_{curve}(\Psi) = \frac{1}{1+\exp\left[(\Psi - P_{50})/s_x\right]}. \quad (9)$$

This reparameterised curve follows the mathematical framework of Pammenter and Van der Willigen (1998) and Duursma and Choat (2017). It gives 50% loss at P50. The magnitude of the PLC slope there is $25/s_x$ percentage points per MPa. The width and P50 remain fixed within each selected hydraulic member; newly produced xylem has the same vulnerability as existing xylem.

Actual fractional loss is now calculated from two state variables:

$$L = 1 - \frac{K_{fun}}{K_{pot}}. \quad (10)$$

K_{pot} is the conductance of all retained xylem if functional; K_{fun} is its actual functional conductance. Both have units L h^{-1} MPa^{-1}.

Initially, potential conductance is $K_0 = K_{pot,0} = 1/R_{xylem,0}$. Initial loss is $L_0 = max\left[L_{initial}, L_{curve}(\Psi_0)\right)$.

Functional conductance starts at $K_{fun,0} = K_0(1 - L_0)$.

The hydraulic resistance becomes

$$R = R_{other} + \frac{1}{max(K_{fun}, \varepsilon_K K_{pot})}. \quad (11)$$

With no growth, turnover or repair, Equations 10 and 11 reduce to the original conductance-loss representation. The numerical floor acts only in the flux calculation; it does not change reported actual conductance or imply survival.

Step order and seasonal xylem addition

Let $\delta = \Delta t_h / 24$ be the internal step in days. Proportional retirement is applied to potential and functional pools first:

$$K_p^r = K_{pot} e^{-\mu\delta}, K_f^r = K_{fun} e^{-\mu\delta}. \quad (12)$$

The rate μ is turnover_rate_per_day. This optional simplification removes functional and damaged xylem in the same proportion, leaving their loss fraction unchanged. It is not an age-cohort or heartwood-development model and defaults to zero.

Growth then adds initially functional capacity:

$$G = K_0 \gamma f_g I_g \delta, K_p^* = K_p^r + G, K_f^* = K_f^r + G. \quad (13)$$

Here γ is growth_rate_per_day, f_g is the forcing growth_factor, and I_g is a binary eligibility gate. The gate equals one only when soil potential and the start-of-substep stem potential are at least growth_soil_min and growth_psi_min, and temperature lies inclusively between growth_temp_min and growth_temp_max. It is zero otherwise. The gate and seasonal factor multiply; warm wet conditions do not trigger growth when $f_g = 0$.

Growth is an absolute conductance addition relative to initial capacity, not exponential compounding, basal-area increment or a carbon-budget calculation. For example, $\gamma = 0.001$ day^{-1} with $f_g = 1$ and an open gate for 150 days adds 15% of initial potential capacity, before turnover and additional embolism. A factor of 0.5 over the same period adds 7.5%.

Optional repair of existing conduits

After growth, the proposed repair increment is

$$J = \left(K_p^* - K_f^*\right)\left[1 - e^{-rI_r\delta}\right), F^* = K_f^* + J\,. \quad (14)$$

The rate r is repair_rate_per_day. The binary repair gate I_r uses its own soil, start-of-substep stem-potential and temperature thresholds. The amount is bounded by the currently nonfunctional capacity. There is no requirement for daylight in this gate. The default rate is zero, so no existing conduits are repaired unless the user deliberately enables this hypothesis.

Growth and repair are distinct processes. Brodribb et al. (2010) provide evidence for growth-associated hydraulic recovery in Callitris. Rehschuh et al. (2020) observed persistent drought-induced embolism in Scots pine over four weeks, whereas Mayr et al. (2020) documented refilling following winter embolism in timberline Norway spruce. These contrasting settings motivate an optional repair mechanism; they do not establish the particular rates or threshold functions above.

Damage is reapplied at the new potential

The implicit hydraulic solve enforces

$$K_{fun,new} = min\left[F^*, K_p^*\left(1 - L_{curve}\left(\Psi_{new}\right)\right)\right), K_{pot,new} = K_p^*. \quad (15)$$

Equivalently,

$$L_{new} = max\left[1 - \frac{F^*}{K_p^*}, L_{curve}\left(\Psi_{new}\right)\right). \quad (16)$$

This is the explicit link between P50 and the recovered state. At P50, actual loss must still be at least 50%; pre-existing damage can make it higher. Under wetter conditions, growth or repair can reduce actual loss. Capacity provisionally restored during a step can be lost again immediately if the new water potential is too negative.

The implementation records gross growth, gross repair, retired potential capacity and new embolism loss separately. Gross repair is not guaranteed to be a net hydraulic gain after Equation 15. Cumulative mortality hazard is never reset by growth, repair or the start of a calendar year.

5 Hydraulic exposure and mortality

Potential based exposure

The default exposure_mode=1 uses the depth of stem potential below P50:

$$x(t_d) = max\left[0, \frac{P_{50} - \Psi(t_d)}{\Psi_{scale}}\right]. \quad (17)$$

The scale is in MPa and makes x dimensionless. With P50 = −3 MPa, stem potential = −4 MPa and scale = 1 MPa, exposure equals 1. With stem potential = −2 MPa, it equals zero.

Alternative exposure based on persistent PLC

Setting exposure_mode=2 replaces Equation 17 with

$$x(t_d) = max\left[0, \frac{L(t_d) - L_{crit}}{1 - L_{crit}}\right]. \quad (18)$$

Here L_{crit} is the fractional threshold loss_threshold. This exposure is between 0 and 1. Because damage can persist, it can sustain excess hazard after rewatering. Growth or enabled repair can subsequently reduce loss and the instantaneous excess hazard. The two modes are alternative hypotheses; the code does not add their effects together.

Duration and survival

$$D_q(t_d) = \int_0^{t_d} x(u)^q \, du, \lambda(t_d) = \lambda_0 + k\, x(t_d)^q. \quad (19)$$

$$H(t_d) = \lambda_0 t_d + k\, D_q(t_d), S(t_d) = e^{-H(t_d)}, p(t_d) = 1 - S(t_d). \quad (20)$$

The standard hazard-survival relationship is described by Klein and Moeschberger (2003) and the Stan survival documentation. Our particular hydraulic exposure and additive hazard are proposed choices to calibrate against mortality observations. They are not established universal hydraulic mortality laws.

The exponent $q > 0$ changes severity weighting. Exposure is integrated in days; both hazard coefficients are in day^{-1}. For an illustrative two days at $x = 1$, $\lambda_0 = 0.0001$ day^{-1} and $k = 0.05$ day^{-1}, cumulative hazard is 0.1002 and mortality probability is about 0.0953. This example is not a fitted prediction.

In mode 1, excess hazard returns to zero when potential rises above P50; accumulated risk remains. In mode 2, persistent PLC can retain excess hazard until functional capacity is restored. Neither mode explicitly simulates a delayed-death compartment; restoration now follows Section 4. Follow-up duration therefore matters, especially in the no-repair PLC formulation.

6 Bayesian inference and uncertainty

Parameter uncertainty

For each particle, the uncertain parameter vector is $\eta = (\lambda_0, k, q, P_{50})$. The code draws independent prior values:

$$\log \lambda_0 \sim N(\log a_0, \sigma_0^2), \log k \sim N(\log a_k, \sigma_k^2). \quad (21)$$

The values a_0 and a_k are prior medians, not means. The exponent and P50 use finite categorical priors specified by q_values, q_prior, p50_offsets and p50_prior. P50 equals the hydraulic base value plus its selected offset. The same P50 controls embolism and mortality exposure. Other hydraulic parameters, including the growth, repair and turnover rates and their eligibility thresholds, and the environmental drivers are fixed. They are not inferred by the Bayesian routine.

Endpoint likelihood

For case j, all N_j trees are alive at its first forcing time. The endpoint number of deaths y_j is modelled as

$$y_j \sim Binomial(N_j, p_j), p_j = 1 - e^{-H_j}. \quad (22)$$

$$\ell(\eta) = \sum_j \left[y_j \log\left(1 - e^{-H_j}\right) - (N_j - y_j) H_j \right] + constant. \quad (23)$$

Deaths occur at unknown times within the observation interval; the other trees survive to its end. The binomial coefficient is omitted because it is constant in the parameters. Conditional independence is assumed within and between cases. Exact event times, repeated inspections, block effects and tree-to-tree frailty are not modelled. Do not enter repeated endpoints from the same cohort as independent cases.

Prior importance sampling

$$p(\eta \mid y) \propto p(y \mid \eta) p(\eta), w_i = \frac{e^{\ell_i - \ell_{max}}}{\sum_r e^{\ell_r - \ell_{max}}}. \quad (24)$$

Particles are drawn from the prior, so normalised likelihoods become their posterior weights. This is self-normalised importance sampling, not MCMC (Owen, 2013). In prior predictive mode, all weights are equal and no survival outcomes update them.

$$ESS = \frac{1}{\sum_i w_i^2}. \quad (25)$$

The code warns if ESS is below the larger of 200 and 5% of the particle count. This heuristic does not prove accuracy: important posterior regions may still have been missed. Repeat with more particles and different seeds. With few observed deaths, several parameter combinations can remain weakly identified.

7 Numerical integration and code structure

Hydraulic step

Environmental drivers are constant over each forcing interval. That interval is subdivided so no internal step exceeds max_step, in hours. A backward Euler update solves

$$C\frac{\Psi_{r+1}-\Psi_r}{\Delta t_h}-Q_{r+1}+E_{r+1}=0. \quad (26)$$

Each substep first applies turnover, gated growth and gated repair to obtain provisional capacities. Their gates use the old stem potential and current interval forcing. Stomatal closure and the P50 damage cap are then evaluated at the new potential during the implicit solve. Provisional capacities remain fixed during that solve. This operator splitting is first order and must be checked for timestep convergence, including near eligibility thresholds. The program uses a sign bracket and bisection, with at most 100 iterations and a relative bracket-width stopping rule. Strong embolism feedback can make the residual nonmonotone; a bracket finds a root but does not establish a unique physically appropriate branch. Timestep convergence is essential near rapid hydraulic deterioration.

The code integrates exposure at internal endpoints:

$$D_{q,r+1}=D_{q,r}+\frac{\Delta t_h}{24}x^q_{r+1}. \quad (27)$$

Endpoint fluxes also accumulate transpiration and uptake volumes. Capacitance C remains constant. Changes in hydraulic capacity do not directly add water to the storage balance, and water/carbon costs of xylem production or refilling are not represented. A consistency check is

$$C\left(\Psi-\Psi_{initial}\right)=V_{uptake}-V_{transpired}. \quad (28)$$

Program organisation

File or procedure	Function
hydraulics.f90	RC physics, vulnerability, fluxes and implicit stepping
soil_potential	Inverse retention curve
fluxes	Conductance, transpiration, resistance and uptake
capacity_update	Retirement, seasonal growth and optional repair
step	Implicit potential solve and P50-constrained functional capacity
mortality.f90	Input handling, exposure integration, inference and outputs
category, normal_draw	Categorical and normal random sampling
summary	Weighted means and discrete empirical quantiles

File or procedure	Function
verify.py	Optional executable regression checks

Each P50 member is simulated for every case. Its P50 remains constant across all years; potential and functional conductance evolve continuously. Exposure is accumulated for every q value, after which mortality particles select the appropriate trajectory and dose. This avoids rerunning the hydraulic model for every mortality particle.

Trajectory and dose storage is approximately $8\left(22+n_q\right)N_{max}N_{case}N_{P50}$ bytes, excluding other arrays. The longest case sets N_{max} for all allocated cases. Very long time series or many cases can therefore require substantial memory.

8 Install compile and run on Linux

Requirements

The model needs a Fortran compiler with Fortran 95 support. GNU Fortran is used in the supplied commands. There are no external Fortran library dependencies. Python 3 is optional and is required only for verify.py. On Ubuntu or Debian, install the tools through your normal software-management process, for example:

```
sudo apt update
sudo apt install gfortran make unzip python3
```

Extract the supplied archive and enter the directory that contains the source files:

```
unzip hydraulic_bayesian_mortality_F95.zip
cd hydraulic_mortality_F95
```

Compile

Compile the hydraulic module before the main program:

```
gfortran -std=f95 -Wall -Wextra -O2 -fcheck=all \
  hydraulics.f90 mortality.f90 -o hydraulic_mortality
```

Alternatively, run make. For additional floating-point diagnostics during testing, add -ffpe-trap=invalid,zero,overflow. Compiler messages and successful compilation do not replace numerical or scientific validation.

Run the supplied prior predictive example

```
./hydraulic_mortality
cat diagnostics.txt
head -n 5 endpoint_predictions.csv
```

The program reads parameters.nml, bayes_settings.nml and cases.csv from the current working directory. Forcing paths are also relative to that directory. Input filenames are fixed; there are no command-line input options.

A successful run prints a completion message, possibly preceded by a low-ESS warning. Outputs listed in Section 13 are replaced on every run. Save each experiment in its own directory, together with its source, inputs, compiler version and results.

The source files are compiled; Markdown files such as README.md and this manual are documentation. They are not Fortran input and should not be passed to the compiler.

9 Prepare forcing and survival cases

Forcing CSV format

Use one header row followed by either five legacy numeric columns or seven extended numeric columns. A file may technically contain either supported row width, but a consistent format is strongly recommended. The first five columns retain their existing order:

Column	Units	Requirements
time_h	Hours	Nonnegative and strictly increasing
T_C	Degrees C	Air temperature
theta	$m^3\ m^{-3}$	Root-zone volumetric moisture, not percent
VPD_kPa	kPa	Nonnegative air VPD
day_factor	Dimensionless	Between 0 and 1
growth_factor	Dimensionless	Optional column 6; seasonal activity in [0,1]
leaf_area_factor	Dimensionless	Optional column 7; total area divided by base leaf_area

```
time_h,T_C,theta,VPD_kPa,day_factor,growth_factor,leaf_area_factor
0.0,15.0,0.35,0.2,0,0.8,1.0
0.5,16.0,0.35,0.3,0,0.8,1.0
1.0,18.0,0.34,0.6,1,0.8,1.0
```

A five-column row implies growth_factor=0 and leaf_area_factor=1; it does not imply year-round growth. Six columns are rejected. Repair can still operate with five columns if its rate is nonzero. Growth and leaf-area factors are externally prescribed; no calendar, phenology, carbon-growth or leaf-shedding model is embedded.

The first row sets the initial output. The second row supplies the drivers over the interval (0.0, 0.5] hours, and the third supplies them over (0.5, 1.0]. There is no interpolation. Initial potential and loss come from parameters.nml, not from the forcing row.

Convert dates to elapsed hours externally. Derive daylight from timestamp and location, or provide a calibrated light factor. There are no built-in sunrise calculations. Correct missing data before running; blank rows, missing fields, nonnumeric tokens, embedded spaces within numeric fields, embedded commas and quoted numeric fields are unsupported. The new reader validates field counts rather than silently ignoring extra values. The parser expects the supplied column order rather than matching header names.

Case CSV format

```
case_id,forcing_file,n_trees,n_dead
control,forcing_control.csv,20,0
```

```
drought,forcing_severe.csv,20,6
heldout,forcing_new.csv,0,0
```

Counts above are illustrative. Use unique IDs without spaces or commas. A positive n_trees supplies endpoint observations when Bayesian updating is enabled. A zero marks prediction-only cases and requires n_dead=0. The code assumes all counted trees were alive at the first row and were assessed at the last row.

Use simple filenames. Quote paths containing a slash, for example "drivers/forcing_A.csv", because unquoted slashes terminate Fortran list-directed input. Avoid spaces and embedded commas in paths. Keep IDs within 64 characters and paths within 256 characters.

Different cases may end at different times. Extend environmental forcing through the actual survival assessment, including post-rewatering follow-up. For tree-specific drivers, use one case per tree with counts 1 and 0 or 1. Repeated drought episodes for the same tree belong in one continuous forcing record, so both conductance pools, damage and cumulative hazard are retained. The program initialises every case independently, does not carry state between runs and has no restart-file mechanism. Use one continuous record for multiple years; do not split the same individual into annual independent cases.

10 Hydraulic parameter reference

All values below are the explicit values in the supplied parameters.nml. They are demonstration values, not validated species traits. The namelist group must begin with &model and end with /.

Code name	Default	Units or interpretation
c	0.5	L MPa^{-1}; constant capacitance
r_other	0.2	MPa h L^{-1}; constant non-xylem resistance
r_xylem	0.8	MPa h L^{-1}; fully functional xylem resistance
height	1.0	m; effective elevation above soil
leaf_area	2.0	m^2; total tree leaf area
pressure	101.325	kPa; atmospheric pressure
gmin	0.002	mol m^{-2} s^{-1}; residual conductance
gmax	0.20	mol m^{-2} s^{-1}; maximum effective conductance
topt	25.0	Degrees C; temperature optimum
twidth	15.0	Degrees C; positive Gaussian width
d0	1.5	kPa; positive VPD-response scale
gs50	−1.5	MPa; stomatal half-response potential
sg	0.25	MPa; positive stomatal-response width

Code name	Default	Units or interpretation
p50	−3.6	MPa; base xylem half-loss potential
sx	0.35	MPa; positive vulnerability width
theta_r	0.05	$m^3\ m^{-3}$; residual soil moisture
theta_s	0.45	$m^3\ m^{-3}$; saturated soil moisture
alpha	3.6	m^{-1}; retention parameter
vn	1.56	Dimensionless; retention exponent n > 1
max_step	0.01	Hours; maximum integration step
initial_psi	−0.2	MPa; shared initial stem potential
initial_loss	0.0	Fraction; initial hydraulic damage
loss_floor	0.000001	Numerical minimum remaining conductance

Growth repair and turnover settings

These entries belong to the same &model namelist. The supplied values are illustrative; repair and turnover are off by default.

Code name	Supplied value	Meaning
growth_rate_per_day	0.001	Initial-capacity fraction added per fully eligible day
turnover_rate_per_day	0.0	Proportional retirement rate per day
growth_soil_min	−0.5	MPa; minimum eligible soil potential
growth_psi_min	−1.0	MPa; minimum eligible old stem potential
growth_temp_min	5.0	Degrees C; inclusive lower growth limit
growth_temp_max	40.0	Degrees C; inclusive upper growth limit
repair_rate_per_day	0.0	Repair rate per day of missing functional capacity
repair_soil_min	−0.2	MPa; minimum eligible soil potential
repair_psi_min	−0.5	MPa; minimum eligible old stem potential
repair_temp_min	5.0	Degrees C; inclusive lower repair limit
repair_temp_max	35.0	Degrees C; inclusive upper repair limit

The parser accepts growth and turnover rates between 0 and 1 day^{-1} and repair rates between 0 and 10 day^{-1}. These numerical bounds are not physiological recommendations. Water-potential thresholds must be nonpositive and each lower temperature limit must be below its upper limit. The output leaf factor is bounded between 0 and 100; zero represents zero leaf area and therefore zero modelled transpiration, including residual leaf water loss. Stem cuticular loss is not separately represented.

Module defaults for the three new rates are zero, so old namelists remain valid. The supplied version 2 parameter file explicitly selects growth rate 0.001. Growth still requires a nonzero seasonal driver. To reproduce the version 1 damage model, set all three rates to zero and use a leaf factor of one. Do not use a calibrated version 1 mortality posterior unchanged after altering hydraulic recovery; recalibrate the coupled model.

Migration from version 1

Existing five-column forcing and old namelists are supported. To isolate the original behaviour, use zero growth, repair and turnover rates and a leaf factor of one. Existing parsers that read trajectory columns by name remain usable; parsers that assume exactly 17 columns must accept the eight appended fields. Scientific-format output now specifies three exponent digits so very small Bayesian weights retain the E marker and can be read by R and Python. All existing output filenames and the posterior/prior mode flag are unchanged.

Constraints and interpretation

Capacitance, xylem resistance, leaf area, pressure, response widths and max_step must be positive. Non-xylem resistance and height must be nonnegative. Conductance must satisfy 0 <= gmin <= gmax; soil parameters require 0 <= theta_r < theta_s <= 1, alpha > 0 and vn > 1.

The model checks 0 <= initial_loss <= 1 and 0 < loss_floor < 1. Initial potential must be nonpositive and greater than −1000 MPa. These numerical limits do not define a physiologically plausible parameter range; use independently supported values.

Keep gs50 distinct from p50. Calibrate resistance and capacitance against flux and potential observations where possible. Stem potential alone cannot separately constrain all hydraulic and conductance parameters.

11 Bayesian parameter reference

The settings file uses the namelist group &bayes, terminated by /. All hazard rates below are per day, although hydraulic stepping uses hours.

Code name	Supplied value	Meaning
exposure_mode	1	1 for P50 deficit; 2 for persistent PLC
psi_scale	1.0	MPa scale for mode 1 exposure
loss_threshold	0.5	Fractional PLC threshold for mode 2
n_particles	8000	Prior importance particles
seed_value	271828	Nonnegative random seed

Code name	Supplied value	Meaning
use_observations	.false.	Enable endpoint Bayesian updating when true
lambda0_median	0.0001	Prior median baseline hazard, day^{-1}
lambda0_logsd	1.5	Natural-log SD of baseline hazard
k_median	0.05	Prior median stress coefficient, day^{-1}
k_logsd	1.0	Natural-log SD of stress coefficient
nq	3	Number of q categories
q_values	1, 2, 3	Allowed exposure exponents
q_prior	About ⅓ each	Relative category probabilities
np50	3	Number of P50 categories
p50_offsets	−0.2, 0, 0.2	MPa offsets from hydraulic p50
p50_prior	0.25, 0.5, 0.25	Relative category probabilities

The example P50 values are therefore −3.8, −3.6 and −3.4 MPa. Both categorical probability arrays are normalised automatically. They describe shared parameter uncertainty, not individual-tree variation. There is no continuous interpolation between categories.

Grid sizes must be 1 to 8. Retained q values must satisfy 0 < q <= 10. Prior weights must be nonnegative with a positive sum, and all effective P50 values must be negative. Particle count is restricted to 100 through 1,000,000. Both log-SDs must be between 0 and 5; zero fixes the associated coefficient. Medians must be positive. The checks require 0 < psi_scale <= 100 and 0 <= loss_threshold < 1.

To fix P50 at the value in parameters.nml, use np50=1, p50_offsets=0.0, p50_prior=1.0. To fix q at 2, use nq=1, q_values=2.0, q_prior=1.0. Specify every retained array entry intentionally; unassigned entries retain module defaults.

Changing the exposure mode or potential scale changes the meaning of k. Reconsider and recalibrate its prior accordingly. A fixed seed is repeatable with the same compiler/runtime and settings; random streams need not match between compilers. The seed routine reduces the input seed modulo 1,000,000, so seeds separated by that amount initialise the same sequence.

12 Calibrate and apply the model

First establish the hydraulic model

Fit the soil retention relation and constrain leaf area independently. Use measured conductance, stem potential and uptake or sap flow to constrain the hydraulic parameters. Basal sap flow represents uptake, which can differ

from instantaneous transpiration while storage changes. Hold out days or drought episodes to test predictions before linking them to mortality.

The current code does not infer C, R or stomatal parameters from observations. Their calibration, including the new recovery rates, must be performed externally. The Bayesian routine only updates the mortality coefficients, the categorical q and the shared categorical P50 using endpoint death counts.

Examine prior predictions

Keep use_observations=.false. and run plausible environmental scenarios. Check whether the resulting risks are compatible with independent survival evidence. The supplied medians, widths and categorical probabilities are illustrative. Use control observations and relevant mortality studies to justify priors rather than treating the defaults as species estimates.

Update using observed outcomes

1. Prepare one continuous forcing record per independent cohort or tree, including recovery and the final survival assessment.

2. Enter real n_trees and n_dead values in cases.csv. Set both counts to zero for held-out prediction cases.

3. Set use_observations=.true. in bayes_settings.nml and rerun.

4. Inspect ESS, maximum weight and the parameter summaries. Repeat with larger particle counts and different seeds to check stability.

5. Assess prediction on independent cases. Observed cases included in the likelihood provide in-sample checks, not independent validation.

Small cohorts, few deaths and similar exposures can leave baseline hazard, severity response and P50 weakly identified. Fix independently known parameters or use justified informative priors. Zero observed deaths can constrain upper risk but does not establish a precise lethal threshold.

Synthetic calibration demonstration

The package contains invented counts of 0/20, 1/20 and 6/20 solely to exercise the inference workflow. They are not observations from the greenhouse experiment. In a disposable example folder:

```
cp cases.csv cases_prior.csv
cp cases_SYNTHETIC_calibration.csv cases.csv
```

Edit bayes_settings.nml to set use_observations=.true., then run the executable. Restore cases_prior.csv and the false setting afterwards if you want the original prior run. The short synthetic forcing uses eight days of drying, two days at minimum moisture and four days after rewetting; it does not reproduce the experimental treatment schedule. It retains five columns, so growth is off in that example.

Two year seasonal example

examples/two_years contains a complete input set with a continuous 730-day record, two synthetic droughts, seasonal growth and prescribed leaf-area changes. It uses seven columns, six-hourly drivers, 2000 particles and a 0.10 h internal step. The coarse environmental sampling is for demonstration; use suitably resolved

measurements for research and test both driver resolution and internal timestep. Time is an illustrative 365-day annual cycle, not a real dated leap-year calendar.

After compilation in the package root:

```
cd examples/two_years
../../hydraulic_mortality
```

Repair and turnover are zero in this example, allowing the growth contribution to be identified. Compare with growth_rate_per_day=0, then test separately justified repair rates. Archive each variant in a different directory. Mortality follows the same surviving cohort across both years and is not reset between seasons.

13 Outputs and interpretation

The program replaces the following files in the current working directory:

Output	Contents and use
hydraulic_trajectories.csv	Environmental drivers and hydraulic states for each case and P50 member
exposure_doses.csv	Cumulative dose for each case, P50 member and q
parameter_particles.csv	Sampled parameters, uniform prior weights and current weights
parameter_summary.csv	Weighted parameter means and 2.5%, 50%, 97.5% quantiles
risk_timeseries.csv	Cumulative mortality probabilities and mean survival through time
endpoint_predictions.csv	Endpoint risk and, where a cohort size exists, replicate death-count intervals
diagnostics.txt	Inference mode, ESS, largest weight and water-balance residual

New conductance and recovery outputs

The eight fields appended to the version 1 trajectory schema are:

Field	Units and interpretation
Kpot_L_h_MPa	Potential retained conductance, L h^{-1} MPa^{-1}
Kfun_L_h_MPa	Actual functional conductance, same units
growth_factor	Supplied seasonal growth activity
leaf_area_factor	Supplied ratio of total to base leaf area
growth_added_K	Cumulative gross capacity added by growth
repair_added_K	Cumulative gross capacity restored by repair
retired_potential_K	Cumulative potential capacity removed by turnover

Field	Units and interpretation
new_damage_K	Cumulative functional capacity removed by new embolism

Cumulative capacity increments have the same conductance units as K, not water-volume units. Potential capacity should equal initial capacity plus growth_added_K minus retired_potential_K. Actual loss should equal one minus Kfun_L_h_MPa/Kpot_L_h_MPa and must be at least the current curve-implied loss. Actual PLC may now decrease, whereas cumulative mortality risk must remain nondecreasing. Gross repair and new damage can both increase in one step when restored capacity is immediately re-embolised.

Hydraulic states and exposure

The first 17 trajectory fields retain their order from version 1. They are case_id, p50_member, P50_MPa, time_h, T_C, theta, VPD_kPa, day_factor, soil_MPa, stem_MPa, PLC_percent, gsw_mol_m2_s, E_L_h, Qin_L_h, R_MPa_h_L, transpired_L and uptake_L. Fluxes are instantaneous forcing-row endpoint values; cumulative volumes use internal integration steps. PLC is reported as a percentage, although loss_threshold is a fraction.

Dose fields are case_id, p50_member, q, time_h and dose_days. Join a particle to dose records using its selected P50 value/member and q, not by assuming that one trajectory is the posterior mean. Raw hydraulic member outputs themselves are not posterior-weighted summaries.

Risk and uncertainty

Risk fields are case_id, inference, time_h, elapsed_days, mean_Pdeath, p025, median, p975 and mean_survival. The inference label is prior_predictive or posterior. Risk starts at zero at the first row, and cumulative risk cannot decrease. Mean survival equals one minus mean mortality probability.

The probability interval describes parameter uncertainty conditional on the model, drivers and fixed hydraulic traits. It is not a confidence interval for a binary individual outcome, and the bands are pointwise rather than simultaneous across time. Additional hydraulic and forcing uncertainty is not included beyond the specified P50 grid.

The replicate-death-count interval adds binomial outcome variability for a new cohort with the same size and environment. Prediction-only cases have no supplied cohort size and receive NA count intervals. The program samples these counts; their quantiles have Monte Carlo error. The prior_weight column stays uniform even in posterior mode.

Read all scientific-notation values as numbers. Leading spaces in formatted CSV fields are normal Fortran output. The diagnostic log mean likelihood omits binomial constants and should not be treated as a fully normalised model evidence for comparing different datasets.

14 Verification troubleshooting and limitations

Run the executable checks

Keep an unmodified copy of the example package and run:

```
python3 verify.py
```

With GNU Fortran installed, verify.py compiles and tests the model in temporary directories. It checks growth, gated repair, turnover, the P50 cap, capacity and water balance, Bayesian weights, baseline survival, malformed inputs and timestep sensitivity. Run python3 tests/check_two_years.py for the seasonal checks. Both scripts use the supplied example parameters and leave working outputs unchanged.

All executable checks passed with GNU Fortran 13.3.0. In the two-year example, maximum water-balance error was 1.16×10^{-8} L. Halving the step from 0.10 to 0.05 h changed prior mean mortality probability by at most 9.99×10^{-4} and stem potential by 0.00804 MPa. The no-recovery comparison reproduced version 1 stem potentials and mean risk. Full settings and results are in VERIFICATION.md. These are numerical checks on synthetic examples, not empirical validation or universal error bounds.

Common problems

Symptom	Action
Compiler cannot find a module	Compile both files together, with hydraulics.f90 first
Cannot open an input file	Run from the input directory and check relative paths
Invalid namelist	Check group name, spelling, commas and terminating slash
Invalid moisture	Use volumetric fractions and check retention bounds
Unexpected time response	Check hours, row-end forcing convention and driver resolution
Posterior mode stops	Supply at least one positive observed cohort size
Prior label remains	Set use_observations=.true. and rerun
Low ESS or one dominant weight	Inspect priors; increase particles and repeat seeds; consider a better sampler
Extreme resistance or collapse	Inspect loss, floor, widths and convergence with smaller steps
Runs require too much memory	Reduce cases or record count, or redesign array storage

Scientific limits and reporting

The single node omits leaf-stem segmentation, multiple soil layers, finite water-store bounds, dynamic soil/root resistance, carbon starvation, pests and explicit xylem age cohorts. Growth and repair are prescribed capacity processes, not a biomass or metabolic model. Height, non-xylem resistance and capacitance remain fixed; updating capacitance requires a consistent water-storage formulation and is deferred. Seasonal leaf area is supplied externally. Do not interpret version 2 as a fully validated tree-development model. The PLC mode retains excess hazard while damage persists; its follow-up horizon must be justified. Neither P50 nor the numerical conductance floor is a universal death threshold.

For a paper, report parameter sources and units, recovery assumptions, gates, priors, initial states, forcing resolution, timestep tests, particles, seed, ESS, follow-up and independent validation. Archive code and inputs. Distinguish the proposed exposure-hazard and recovery rules from their hydraulic and survival-analysis foundations.

15 References and equation provenance

Hydraulic and stomatal foundations

Campbell, G. S., and Norman, J. M. (1998). *An introduction to environmental biophysics*. 2nd edition. Springer. https://doi.org/10.1007/978-1-4612-1626-1. Standard biophysical transport reference relevant to Equation 8 and its assumptions.

Duursma, R. A., and Choat, B. (2017). fitplc - an R package to fit hydraulic vulnerability curves. *Journal of Plant Hydraulics*, 4, e002. https://doi.org/10.20870/jph.2017.e002. Supports vulnerability-curve fitting, P50, slope and uncertainty; relevant to Equation 9.

Jarvis, P. G. (1976). The interpretation of the variations in leaf water potential and stomatal conductance found in canopies in the field. *Philosophical Transactions of the Royal Society of London B*, 273, 593-610. https://doi.org/10.1098/rstb.1976.0035. Framework for environmental stomatal response functions; our Equation 5 is an adaptation.

Pammenter, N. W., and Van der Willigen, C. (1998). A mathematical and statistical analysis of the curves illustrating vulnerability of xylem to cavitation. *Tree Physiology*, 18, 589-593. https://doi.org/10.1093/treephys/18.8-9.589. Mathematical foundation for sigmoidal vulnerability curves and their position and slope parameters.

Ruffault, J., Pimont, F., Cochard, H., Dupuy, J.-L., and Martin-StPaul, N. (2022). SurEau-Ecos v2.0: a trait-based plant hydraulics model for simulations of plant water status and drought-induced mortality at the ecosystem level. *Geoscientific Model Development*, 15, 5593-5626. https://doi.org/10.5194/gmd-15-5593-2022. Supports the compartment water-balance framework and remaining-conductance relationship; our model is a substantial simplification.

Tuzet, A., Perrier, A., and Leuning, R. (2003). A coupled model of stomatal conductance, photosynthesis and transpiration. *Plant, Cell & Environment*, 26, 1097-1116. https://doi.org/10.1046/j.1365-3040.2003.01035.x. Related water-potential-dependent stomatal control; the full published model is not implemented.

van Genuchten, M. Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44, 892-898. https://doi.org/10.2136/sssaj1980.03615995004400050002x. Direct foundation for the retention relation inverted in Equations 3 and 4.

Zhang, Z., Guan, H., Veneklaas, E., Singha, K., and Batelaan, O. (2025). Revealing seasonal plasticity of whole-plant hydraulic properties using sap-flow and stem water-potential monitoring. *Hydrology and Earth System Sciences*, 29, 3975-3992. https://doi.org/10.5194/hess-29-3975-2025. Related whole-plant RC representation and use of stem potential and sap flow to constrain properties.

Evidence informing recovery choices

Brodribb, T. J., Bowman, D. J. M. S., Nichols, S., Delzon, S., and Burlett, R. (2010). Xylem function and growth rate interact to determine recovery rates after exposure to extreme water deficit. *New Phytologist*, 188, 533-542. https://doi.org/10.1111/j.1469-8137.2010.03393.x. Growth-associated recovery evidence; not a source for the numerical rate chosen here.

Rehschuh, R., Cecilia, A., Zuber, M., Faragó, T., Baumbach, T., Hartmann, H., Jansen, S., Mayr, S., and Ruehr, N. (2020). Drought-induced xylem embolism limits the recovery of leaf gas exchange in Scots pine. *Plant Physiology*, 184, 852-864. https://doi.org/10.1104/pp.20.00407. Evidence for persistent drought embolism and the need to distinguish rehydration from recovery of conductance.

Mayr, S., Schmid, P., Beikircher, B., Feng, F., and Badel, E. (2020). Die hard: timberline conifers survive annual winter embolism. *New Phytologist*, 226, 13-20. https://doi.org/10.1111/nph.16304. Evidence for refilling in a winter-embolism setting; it does not justify a universal drought-refilling rate.

Statistical foundations

Klein, J. P., and Moeschberger, M. L. (2003). *Survival analysis: Techniques for censored and truncated data*. 2nd edition. Springer. https://doi.org/10.1007/b97377. General hazard, survival and censoring framework; it does not establish the hydraulic exposure function.

Owen, A. B. (2013). *Monte Carlo theory, methods and examples*. Chapter 9, Importance sampling. https://artowen.su.domains/mc/Ch-var-is.pdf. Supports self-normalised importance sampling and weight-based ESS; relevant to Equations 24 and 25.

Stan Development Team (accessed 3 November 2026). *Stan User's Guide*, Survival models. https://mc-stan.org/docs/stan-users-guide/survival.html. Explains cumulative hazard and survival, censoring likelihoods and Bayesian examples. The Fortran program does not use Stan.

Attribution boundaries

Equations 17 to 19 define the exposure-hazard hypothesis used here. The exact P50 deficit, normalised PLC alternative, chosen exponent prior and illustrative lognormal priors are implementation choices. Equations 10 and 12 to 16 define the conductance-pool extension. The growth/repair rates, binary eligibility gates, proportional turnover and common vulnerability of old and new xylem are explicit modelling assumptions. The no-repair case is recovered by setting the repair rate to zero; the original no-restoration model also needs zero growth and turnover. The Gaussian temperature response and particular VPD and logistic stomatal functions are empirical parameterisations. These choices require calibration and testing; they should not be attributed as a complete published model to any one of the references above.